Objective-Driven Sequential Bayesian Experimental Design for Power-Grid Dynamic Learning and Control

[TODO: Authors and affiliations]

***Abstract*—This paper studies sequential Bayesian experimental design for uncertain power-grid dynamics. At each stage, a policy selects a probing bus and duration, observes a noisy frequency response, and updates uncertainty over machine-specific parameters. We distinguish parameter-oriented expected information gain (EIG) from objective-driven mean objective cost of uncertainty (MOCU) for a downstream safety-aware control decision. Random, fixed, myopic, deep adaptive design (DAD), and reinforcement-learning sequential Bayesian optimal experimental design (RL-sBOED), and Step-DAD are evaluated under common simulation banks and held-out protocols. A susceptible–infected–recovered (SIR) model provides an implementation benchmark; IEEE 9-bus, IEEE 14-bus, and IEEE 30-bus systems assess the power-grid application and scalability. [TODO: Add final quantitative findings.]**

***Index Terms*—Bayesian experimental design, power-system dynamics, adaptive design, mean objective cost of uncertainty, information gain.**

I. Introduction

UNCERTAINTY in power-system inertia and fast frequency response complicates frequency-security assessment and control selection. Controlled probing can reduce uncertainty, but an experiment must determine where to probe, how long to probe, and whether later probes should depend on earlier observations. [TODO: Add motivation and citations.]

EIG values parameter learning, but parameter information does not necessarily improve a final control decision. We therefore also study MOCU, which values an experiment by its reduction of expected decision penalty.

The contributions are:

- sequential bus–duration BOED for uncertain power-grid dynamics;
- a safety-aware MOCU objective linking probing to terminal control;
- a matched comparison of random, fixed, myopic, DAD, RL-sBOED, and Step-DAD; and
- SIR, IEEE9, IEEE14, and IEEE30 studies of objective choice, adaptation, horizon, and scalability.

II. Related Work

Summarize BOED and EIG, adaptive and non-myopic design, DAD, RL-sBOED, Step-DAD, MOCU, and power-grid dynamic identification. End with the gap: objective-driven sequential design over power-grid probing buses and durations. [TODO: Add verified references.]

[TODO: Funding and corresponding-author information.]

III. Problem Formulation

A. Dynamic Model and Uncertain Parameters

Let $\mathcal{N} = \{1, \ldots, N\}$ be the set of all physical buses. Partition it as

$$\mathcal{N} = \mathcal{N}_{\mathrm{d}} \cup \mathcal{N}_{\ell}, \qquad \mathcal{N}_{\mathrm{d}} \cap \mathcal{N}_{\ell} = \varnothing, \tag{1}$$

where $\mathcal{N}_{\mathrm{d}} = \{d_1, \ldots, d_{N_{\mathrm{d}}}\}$ contains the N_{d} dynamic-machine buses and $\mathcal{N}_{\ell}$ contains the non-dynamic buses. A physical bus belongs to the probe space whether or not it belongs to $\mathcal{N}_{\mathrm{d}}$. Dynamic states and machine parameters, however, exist only at buses in $\mathcal{N}_{\mathrm{d}}$.

Indexing the reduced dynamic machines by $r = 1, \ldots, N_{\mathrm{d}}$, the general reduced swing model is

$$\dot{\delta}_r(s) = \omega_r(s), \tag{2}$$

$$\begin{aligned} M_r \dot{\omega}_r(s) = P_{m,r} &- \sum_{q=1}^{N_{\mathrm{d}}} B^{\mathrm{red}}_{rq} \sin(\delta_r(s) - \delta_q(s)) \\ &- \left(\frac{K_r}{2\pi} + D_r\right) \omega_r(s) + v_r(s), \end{aligned} \tag{3}$$

where B^{red} is the Kron-reduced coupling matrix and $v_r(s)$ is the external input received by dynamic machine r. The uncertain parameter is

$$\boldsymbol{\theta} = \left(M_1, \ldots, M_{N_{\mathrm{d}}}, K_1, \ldots, K_{N_{\mathrm{d}}}\right) \in \Theta, \qquad \boldsymbol{\theta} \sim p_0(\boldsymbol{\theta}). \tag{4}$$

Thus, there are N possible physical probe locations but only N_{d} pairs (M_r, K_r). In particular, no M or K is assigned to a bus in $\mathcal{N}_{\ell}$. For IEEE9, $(N, N_{\mathrm{d}}) = (9, 3)$; for IEEE14, $(N, N_{\mathrm{d}}) = (14, 5)$.

To show explicitly how all N buses enter the reduced model, partition the full-network susceptance Laplacian according to (1):

$$L = \begin{bmatrix} L_{\mathrm{dd}} & L_{\mathrm{d}\ell} \\ L_{\ell\mathrm{d}} & L_{\ell\ell} \end{bmatrix}. \tag{5}$$

Let $\mathbf{c}_b \in \mathbb{R}^{N_{\mathrm{d}}}$ map an injection at physical bus b to the dynamic-machine equations. The simulator uses

$$\mathbf{c}_b = \begin{cases} \mathbf{e}_r, & b = d_r \in \mathcal{N}_{\mathrm{d}}, \\ -L_{\mathrm{d}\ell} L_{\ell\ell}^{-1} \mathbf{e}_q, & b = \ell_q \in \mathcal{N}_{\ell}, \end{cases} \tag{6}$$

where $\mathbf{e}_r$ and $\mathbf{e}_q$ select a dynamic or non-dynamic bus within its respective partition. Equation (6) is the key distinction between probing a machine bus and probing a load bus.

B. Bus–Duration Probe Space

A diagnostic action is

$$\xi_t = (b_t, \tau_t) \in \Xi = \mathcal{N} \times \mathcal{T}, \tag{7}$$

where b_t may be any of the N physical buses and τ_t is the pulse duration. The amplitude A is fixed across actions. The injected Hann pulse is

$$P_{\mathrm{probe}}(s;\tau_t) = \begin{cases} \frac{A}{2}\left[1-\cos\left(\frac{2\pi s}{\tau_t}\right)\right], & 0 \le s \le \tau_t, \\ 0, & s > \tau_t. \end{cases} \tag{8}$$

During a diagnostic experiment, $u_{\mathrm{ctrl}} = 0$ and the input vector in (3) is

$$\mathbf{v}^{p}(s; b_t, \tau_t) = \mathbf{c}_{b_t} P_{\mathrm{probe}}(s;\tau_t). \tag{9}$$

Consequently, a probe at dynamic bus $b_t = d_r$ gives

$$\mathbf{v}^{p}(s; d_r, \tau_t) = \mathbf{e}_r P_{\mathrm{probe}}(s;\tau_t), \tag{10}$$

whereas a probe at non-dynamic bus $b_t = \ell_q$ gives

$$\mathbf{v}^{p}(s; \ell_q, \tau_t) = -L_{\mathrm{d}\ell} L_{\ell\ell}^{-1} \mathbf{e}_q P_{\mathrm{probe}}(s;\tau_t). \tag{11}$$

Thus, a load-bus probe affects the dynamic machines through the network map; it does not require a fictitious load-bus swing equation.

The terminal-control simulation is separate from the probe simulation. The probe is set to zero, a configured contingency is applied at dynamic index r_c, and supplementary active-power control is applied at dynamic index r_u:

$$\begin{aligned} \mathbf{v}^{c}(s; u_{\mathrm{ctrl}}) = \mathbf{e}_{r_c} P_{\mathrm{cont}}(s) + \mathbf{e}_{r_u} u_{\mathrm{ctrl}} h_{\mathrm{ctrl}}(s), \\ P_{\mathrm{probe}} = 0. \end{aligned} \tag{12}$$

where h_{ctrl} is the configured step, Hann, or ramp profile. Equations (9)–(12) therefore cover the general probe, dynamic-bus probe, non-dynamic-bus probe, and operational-control cases without assigning dynamics to load buses.

With six durations, IEEE9 has $9 \times 6 = 54$ actions and IEEE14 has $14 \times 6 = 84$ actions. Exact action repetition is prohibited within one T-step rollout.

The complete experimental history before stage t is

$$h_{t-1} = \big((\xi_1, \mathbf{y}_1), \ldots, (\xi_{t-1}, \mathbf{y}_{t-1})\big), \tag{13}$$

and an adaptive policy selects

$$\xi_t \sim \pi_t(\cdot \mid h_{t-1}), \qquad t = 1, \ldots, T. \tag{14}$$

The common horizon T is a fixed probe budget, not a term in the operational cost. A Fixed policy chooses $\xi_{1:T}$ before observing data; an adaptive policy may change later actions as a function of the realized history.

C. Observation and Belief Update

Let $\Delta f_o(s; \boldsymbol{\theta}, \xi_t) = \omega_o(s)/(2\pi)$ denote the frequency deviation at the fixed observation bus. At prespecified indices $q_1, \ldots, q_{N_{\mathrm{obs}}}$, the clean observation is

$$\mathbf{g}(\boldsymbol{\theta}, \xi_t) = \left[\Delta f_o(s_{q_1}), \ldots, \Delta f_o(s_{q_{N_{\mathrm{obs}}}})\right]^{\top}. \tag{15}$$

The measured vector satisfies

$$\mathbf{y}_t = \mathbf{g}(\boldsymbol{\theta}^{\star}, \xi_t) + \boldsymbol{\epsilon}_t, qquad \boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \sigma_y^2 \mathbf{I}). \tag{16}$$

The principal power-grid experiments use $N_{\mathrm{obs}} = 5$. Conditional independence across reset-to-equilibrium probes gives

$$p(\boldsymbol{\theta} \mid h_t) \propto p_0(\boldsymbol{\theta}) \prod_{k=1}^{t} p(\mathbf{y}_k \mid \boldsymbol{\theta}, \xi_k). \tag{17}$$

For particles $\{\boldsymbol{\theta}^{(n)}, w_n^{(0)}\}_{n=1}^{N}$, the implemented update is

$$w_n^{(t)} = \frac{w_n^{(t-1)} p(\mathbf{y}_t \mid \boldsymbol{\theta}^{(n)}, \xi_t)}{\sum_{m=1}^{N} w_m^{(t-1)} p(\mathbf{y}_t \mid \boldsymbol{\theta}^{(m)}, \xi_t)}. \tag{18}$$

D. Parameter-Oriented Objective

For EIG experiments, the terminal utility of policy π is

$$J_{\mathrm{EIG}}(\pi) = \mathbb{E}_{\boldsymbol{\theta}^{\star}, H_T \mid \pi}\left[D_{\mathrm{KL}}(p(\boldsymbol{\theta} \mid H_T) \| p_0(\boldsymbol{\theta}))\right], \tag{19}$$

which is maximized. EIG measures parameter learning and does not directly encode the downstream control decision.

E. Terminal Safety Requirement

Probe simulations and terminal-control simulations are separate. During a probe, the terminal control is zero; during terminal evaluation, the probe is zero. For parameter $\boldsymbol{\theta}$ and candidate supplementary active power injection $u \in \mathcal{U}$, let $r(\boldsymbol{\theta}, u)$ be maximum absolute ROCOF and $f_{\min}(\boldsymbol{\theta}, u)$ be minimum frequency deviation under the fixed contingency. Safety is

$$\mathsf{Safe}(\boldsymbol{\theta}, u) \iff r(\boldsymbol{\theta}, u) \le r_{\max} \;\wedge\; f_{\min}(\boldsymbol{\theta}, u) \ge \underline{f}. \tag{20}$$

The model-specific minimum safe control, computed offline on the ordered grid $\mathcal{U}$, is

$$U(\boldsymbol{\theta}) = \min\{u \in \mathcal{U} : \mathsf{Safe}(\boldsymbol{\theta}, u)\}. \tag{21}$$

The bank stores $U_n = U(\boldsymbol{\theta}^{(n)})$ for every particle.

Given posterior weights $\mathbf{w}^{(T)}$, the deployed control is the posterior $(1-\alpha)$ quantile of U, plus margin m, rounded upward to the admissible grid:

$$u_T = \mathrm{ceil}_{\mathcal{U}}\left[Q_{1-\alpha}\left(U \mid \mathbf{w}^{(T)}\right) + m\right]. \tag{22}$$

Here Q_q is the smallest value whose weighted empirical cumulative probability is at least q. The principal MOCU experiments use $\alpha = 0.05$, $m = 0$, and therefore target 95% posterior safety.

F. Safety-Aware OCU and MOCU

For a system requiring control U, define the realized cost

$$L(u, U) = u + \lambda(U - u)_+ + \kappa \mathbf{1}\{U > u\}, \tag{23}$$

where $(x)_+ = \max(x, 0)$, λ penalizes under-control magnitude, and κ optionally penalizes a safety violation. The model-specific optimum has cost $L(U, U) = U$. Hence

$$\mathrm{OCU}(\boldsymbol{\theta}, h_T) = L(u_T, U(\boldsymbol{\theta})) - U(\boldsymbol{\theta}), \tag{24}$$

TABLE I
COMPARED METHODS

Method	Adaptive	Non-myopic	Learned
Random	No	No	No
Fixed	No	Open-loop	No
Myopic	Yes	No	No
DAD	Yes	Yes	Yes
RL-sBOED	Yes	Yes	Yes
Step-DAD	Yes	Yes	Semi-amortized

and belief MOCU is

$$\mathrm{MOCU}(h_T) = \sum_{n=1}^{N} w_n^{(T)} \left[L(u_T, U_n) - U_n \right]. \tag{25}$$

The implementation uses $\lambda = 1/\alpha = 20$ and $\kappa = 0$. This Bayes-loss alignment makes the minimizer of expected (23) the $(1-\alpha)$ quantile used in (22); the terminal rule and reported loss therefore optimize the same decision problem.

Finally, the objective-driven sequential design problem is

$$\pi^\star_{\mathrm{MOCU}} = \arg\min_{\pi} \mathbb{E}_{\boldsymbol{\theta}^\star, H_T|\pi} \left[\mathrm{MOCU}(H_T) \right], \quad \text{subject to } |H_T| = T. \tag{26}$$

The expectation includes the unknown true system, observation noise, and any policy randomization. Lower terminal MOCU is better. This objective values only information that improves the downstream control decision; two posteriors may have different parameter entropy but identical MOCU if they imply the same control-relevant distribution of U.

IV. Compared Design Methods

Random ignores observations. Fixed selects an open-loop sequence. Myopic optimizes immediate expected utility after each observation. DAD learns an amortized history-conditioned policy offline. RL-sBOED learns a sequential policy with stepwise rewards. [TODO: Give reproducible implementation details.] Step-DAD starts from an amortized DAD policy and performs limited history-specific online refinement. Report its offline DAD training cost, online refinement cost, and action-selection cost separately. [TODO: Describe the implementation and deviations from the original paper.]

V. Experimental Setup

A. SIR EIG Benchmark

SIR verifies the EIG implementation on a simpler nonlinear model; it is not the application contribution. [TODO: State its parameters, actions, and noise.]

B. IEEE9 Experiments

IEEE9 has nine physical probe buses, three dynamic machines, and six uncertain M_i/K_i coordinates. Six durations give 54 actions. Evaluate EIG and MOCU with $N_{\mathrm{obs}} = 5$, $\sigma = 0.005$, $T = 3, \ldots, 7$, and seeds 101, 202, and 303. [TODO: Insert final durations and priors.]

C. IEEE14 MOCU Extension

IEEE14 has 14 physical probe buses, five dynamic machines, and ten uncertain M_i/K_i coordinates. Six durations give 84 actions. The MOCU workflow, methods, noise, horizons, and seeds match IEEE9; only IEEE-system physics and machine parameters differ. [TODO: Insert the validated durations.]

D. IEEE30 MOCU Scalability Study

IEEE30 extends the same objective-driven MOCU protocol to a larger network. Use machine-specific uncertain parameters only for its dynamic machines, allow probing at all configured physical buses, and retain the same methods, noise, horizons, seeds, terminal objective, and held-out evaluation rules used for IEEE9 and IEEE14. [TODO: Insert the verified dynamic-machine mapping, prior ranges, action count, bank size, and audited six-duration combination.]

E. Duration Selection and Evaluation

Select six durations before policy training using a method-independent search over 281 values from 0.20 to 3.00 s. Screen adaptive value, non-myopic value, and observation-dependent branching; do not tune durations for a reported method. All methods share physical banks, held-out systems, and noise realizations. Report three-seed mean and sample standard deviation, plus paired held-out differences. [TODO: Add bank sizes and evaluation sample counts.]

VI. Results

A. SIR Implementation Validation

[TODO: Report EIG and runtime for every method.]

B. IEEE9 EIG Results

[TODO: Report mean $\pm$ standard deviation for $T = 3, \ldots, 7$.]

C. IEEE9 MOCU Results

[TODO: Report terminal MOCU, safety, paired differences, and controls.]

D. IEEE14 MOCU Results

[TODO: Report the same methods and metrics as IEEE9.]

E. IEEE30 MOCU Results

[TODO: Report the same methods and metrics as IEEE9 and IEEE14, emphasizing scalability in action count, latent dimension, training time, and online time.]

F. Computational Cost

[TODO: Separate bank generation, offline training, online selection, and Step-DAD history-specific refinement time.]

TABLE II
TERMINAL MOCU ACROSS THREE SEEDS (MEAN ± STANDARD DEVIATION)

Method	$T = 3$	$T = 4$	$T = 5$	$T = 6$	$T = 7$
Random	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
Fixed	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
Myopic	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
DAD	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
RL-sBOED	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
Step-DAD	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]

VII. DISCUSSION

Discuss when adaptation and non-myopic planning improve the downstream objective, why Fixed can be competitive, why EIG and MOCU can disagree, and how horizon and system size affect performance. Limitations include simulated Kron-reduced networks, prescribed priors, fixed probe amplitude, discrete actions, simplified terminal control, and three training seeds. Machine-specific K_i allocations introduced for the larger IEEE systems are modeling assumptions unless directly supported by a cited dynamic-data source.

VIII. CONCLUSION

This paper formulates power-grid probing as sequential BOED and connects learning to safety-aware terminal control through MOCU. [TODO: State only final conclusions supported by completed three-seed results.]

ACKNOWLEDGMENT

[TODO: Add funding and computing resources.]

REFERENCES

[1] [TODO: Add verified references for BOED, DAD, RL-sBOED, Step-DAD, MOCU, and power-grid dynamic identification.]